BIM308 – Web Server Programming – 1st Assignment

In this assignment, you will develop a simple ASP.NET Core MVC application to understand:

- The MVC (Model–View–Controller) structure
- The role of Models, Views, and Controllers
- Passing data from Controller to View
- Razor syntax basics
- Default MVC routing

1. Creating the Project

Create a new **ASP.NET Core Web App (Model-View-Controller)** project.
Name the project **BookLibrary**.
Run the application to ensure it builds and works correctly.
Examine the project structure and identify:

- The Models folder
- The Views folder
- The Controllers folder
- The default controller and default action

2. Creating the Model

Inside the **Models** folder, create a class named **Book**.
Add:

- An integer property for the book's ID (**BookID**).
- A string property for the book title (**Title**).
- A string property for the book's director (**Director**).
- A string array property for the 5-star rating review (**Stars**).
- An integer property for the release year (**ReleaseYear**).
- A string property for the book's image url (**ImageUrl**).
- A double property for the book's price (**Price**).

Do not add any methods.

3. Creating the Controller

Inside the **Controllers** folder, create a controller named **LibraryController**.
Add an action method named **Index**.
Inside the Index action:

- Create a list containing at least **five** books with sample data.

Send the list to the View.

4. Creating the View

Inside the **Views** folder, create a new folder named **Library**.
Inside this folder, create a Razor View named **Index**.
Make the view strongly typed using a list of **Book**.
Display the following (**you are free to display whatever information you want here**):

- A book's image
- Rating info
- Price info

Use Razor syntax to iterate over the list. Each book's information must be displayed.

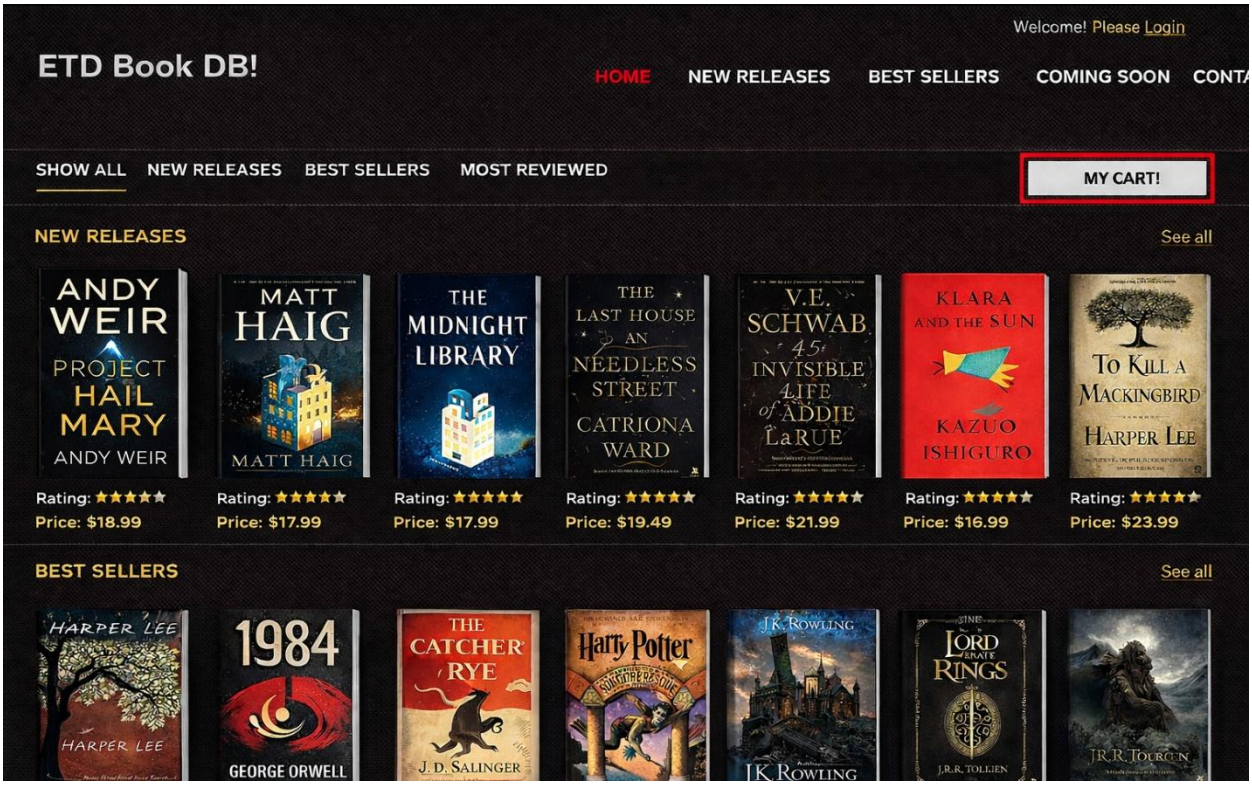
5. Routing

Use the default MVC routing configuration.
Run the application.
Access the page using the route.
Verify that the page loads correctly.
Test /Library and observe the result.

6. Execution

When the application runs:

- The page opens without errors.
- The book's information and description is visible.
- At least five books are listed.
- The routing works correctly.



An Example View

Submitting the Assignment

Zip up your project which should be named as **yourlastname_yourID_hw1.zip** and submit it to ESTUOYS platform until **March 10 at 23:59**.

Only one group member should make submission. Also, please provide a **ReadMe.txt** file that gives information about the group members. Otherwise, your colleague might not receive any points.

Do not forget to put your Project into a container by using the **Dockerfile** that is given to you.

Late submission is not allowed by the system. Submissions via e-mail will not be accepted by the instructors and they will be deleted immediately.